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Ontogenetic variation in salinity tolerance and ecophysiology of coastal dune plants

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- **Background and Aims** Global climate change includes shifts in temperature and precipitation, increases in the frequency and intensity of extreme weather events and sea level rise, which will drastically impact coastal ecosystems. The aim of this study is to quantify salinity tolerance and to identify physiological mechanisms underlying tolerance across wholeplant ontogeny in two widespread native coastal plant species in Hawai'i, *Jacquemontia sandwicensis* (Convolvulaceae) and *Sida fallax* (Malvaceae).
- **Methods** At the seed, seedling, juvenile and mature ontogenetic stages, plants were exposed to high salinity watering treatments. Tolerance was assayed as the performance of stressed compared with control plants using multiple fitness metrics, including germination, survival, growth and reproduction. Potential physiological mechanisms underlying salinity tolerance were measured at each ontogenetic stage, including: photosynthesis and stomatal conductance rates, leaf thickness, leaf mass per area and biomass allocation.
- **Key Results** Salinity tolerance varied between species and across ontogeny but, overall, salinity tolerance increased across ontogeny. For both species, salinity exposure delayed flowering. Physiological and morphological leaf traits shifted across plant ontogeny and were highly plastic in response to salinity. Traits enhancing performance under high salinity varied across ontogeny and between species. For *J. sandwicensis*, water use efficiency enhanced growth for juvenile plants exposed to high salinity, while chlorophyll content positively influenced plant growth under salinity in the mature stage. For *S. fallax*, transpiration enhanced plant growth only under low salinity early in ontogeny; high transpiration constrained growth under high salinity across all ontogenetic stages.
- **Conclusions** That salinity effects vary across ontogenetic stages indicates that demographic consequences of sea level rise and coastal flooding will influence population dynamics in complex ways. Furthermore, even coastal dune plants presumably adapted to tolerate salinity demonstrate reduced ecophysiological performance, growth and reproduction under increased salinity, highlighting the conservation importance of experimental work to better project climate change effects on plants.

Key Words: Coastal dunes, ecophysiology, ontogeny, salt tolerance, salt stress, plant functional traits, phenotypic plasticity.

INTRODUCTION

Climate change factors such as rising sea levels, changes in precipitation regimes and storm frequency will increase salinity exposure for coastal areas worldwide (Nicholls and Cazenave, 2010; IPCC, 2014). Coastal ecosystems provide important ecological services, including protection from erosion, buffering storms and critical habitat for plant species, nesting sea birds, marine mammals and sea turtles (UNEP, 2006; Martínez *et al.*, 2007; Duarte *et al.*, 2013), and their stability during climate change is critical. Although it seems likely that coastal plants have adapted tolerance to saline conditions, high salinity has been observed to limit recruitment, shift phenology and alter plant species distributions in a variety of coastal ecosystems, including wetlands (Spencer *et al.*, 2016; Middleton, 2016; Ury *et al.*, 2019), mangrove forests (Gilman *et al.*, 2008) and sand dunes (Feagin *et al.*, 2005; Greaver and Sternberg, 2007). Furthermore, simulations of acute seawater flooding have reported reduced reproductive output and altered maternal effects (Hanley *et al.*, 2019a) and

effects that cascade up to the population level variation via plant salt sensitivity (White *et al.*, 2014). Although similar responses have been detected among congeners (Hanley *et al.*, 2019b), with few exceptions (Suárez, 2011; Pereira *et al.*, 2019), little is known about the impact of long-term salinity exposure across the broader diversity of supralittoral plant species. Additional research is needed to characterize species tolerance to high salinity in order to better understand why some species are more vulnerable than others. This information will facilitate conservation of these critical habitats and provide insights into the evolutionary ecology of coastal plants.

Salinity tolerance is highly variable across plant species due to differences in the sensitivity of physiological performance and trait expression to high salinity (Flowers *et al.*, 1977; Greenway and Munns, 1980; Munns and Tester, 2008), ranging from species with complete salinity intolerance to halophytes, rare specialists that tolerate high levels of salinity and perform even better under high salinity than under non-saline conditions

(Flowers *et al.*, 2010). Highly tolerant plants are able to avoid salt ion uptake, or sequester salt ions into leaf vacuoles, increase concentration of compatible solutes in the cytoplasm (Munns and Tester, 2008) and maintain a balanced cytosolic Na⁺/K⁺ ratio (Flowers and Colmer, 2015). For sensitive species, salinity exposure alters ion balance, leading to reductions in photosynthesis, protein synthesis and metabolic processes, and ultimately constraining growth (Parida and Das, 2005), although plants can often recover from short-term exposure to high salinity by maintaining water balance via trait plasticity to osmotic stress (Munns, 2002). Although salinity responses are highly species specific, in general, completely tolerant plants exhibit high water use efficiency (WUE; Parida and Das, 2005), high leaf mass per area (LMA) (Munns, 2005), regulated transpiration rates (Harris *et al.*, 2010) and consistent or enhanced growth (Munns and Tester, 2008) under high salinity.

While salinity tolerance and associated traits have been well studied, whether these ecophysiological processes shift across plant development is unclear. This is an important oversight because, in general, stress tolerance and the expression of functional traits change across plant ontogenetic stages, due to both external (i.e. changing environmental conditions as plants grow) and internal (i.e. shifts in biomass allocation and developmentally regulated trait expression) factors. While the link between plant performance and abiotic stress across ontogeny has been well studied for temperature and drought (Wahid *et al.*, 2007; Niinemets, 2010; Ribeiro and Borghetti, 2014), with few exceptions (Houle *et al.*, 2001; Debez *et al.*, 2004), most salinity stress studies are restricted to a single ontogenetic stage. Comparing results across studies on single stages suggests that salinity tolerance may vary across ontogeny. In early ontogeny, high salinity can completely inhibit or delay germination, although seeds of tolerant species may remain viable until salinity levels are reduced and conditions are more favourable for germination (Noe and Zedler, 2000; Vicente *et al.*, 2004; Xiao *et al.*, 2016; Santo *et al.*, 2017). Seedlings, which have limited stored resources and shallow roots, tend to be highly sensitive to salinity stress (Ungar, 1991; Cordeiro *et al.*, 2014), and high salinity at the mature stage has been observed to reduce growth rate, delay flowering and reduce seed set in various species (Houle *et al.*, 2001; Van Zandt *et al.*, 2003; Boscaiu *et al.*, 2005). From these patterns, it seems that salinity tolerance is fairly low early in ontogeny, with an increase from the juvenile to mature ontogenetic stages. However, there is likely to be variation among species, and characterizing the entire ontogenetic trajectory within species will help elucidate these patterns and can provide insights into the life stages most vulnerable to high salinity, which is critical for predicting population resilience under changing climatic conditions.

Hawai'i is an ideal place to test the effects of increasing salinity on coastal plant species. As with other island species, native Hawaiian plants are particularly vulnerable to the impacts of climate change due to reduced species abundances, limited habitat and low genetic variation (Fortini *et al.*, 2013; Harter *et al.*, 2015). As sea levels rise, flooding, salt water intrusion and coastline recession will drastically impact coastal areas across the Hawaiian archipelago (Rotzoll and Fletcher, 2013; Kane *et al.*, 2014; Anderson *et al.*, 2015). Additionally, increased drought stress is projected for the Hawaiian Islands

(Chu *et al.*, 2010; Chen and Chu, 2014; Timm *et al.*, 2015; Zhang *et al.*, 2016), potentially leading to salt accumulation in coastal habitats. These projected changes will result in increased salinity for coastal ecosystems, as well as a loss of habitat. The few previous studies examining salinity tolerance in Hawaiian coastal plants have reported them to be sensitive to substrate salinity and salt spray, particularly at the seedling stage and over long-term exposure (Goldstein *et al.*, 1996; Santiago *et al.*, 2000; Suárez, 2011).

We investigated the effects of salinity on two co-occurring and widespread coastal dune species, *Jacquemontia sandwicensis* (Convolvulaceae) and *Sida fallax* (Malvaceae), commonly found on coasts across the Hawaiian Islands (Warshauer *et al.*, 2009). The aims of this study were to test how salinity affects plant performance across ontogeny, with tolerance defined as the ability of plants to survive, grow and reproduce in high vs. low salinity treatments, and to identify traits linked with salinity tolerance among a suite of morphological and physiological traits, including LMA, photosynthesis, stomatal conductance and WUE. Using relatively long salinity treatments (3 weeks), our experimental approach seeks to mimic extended exposure to salinity as would occur due to saltwater intrusion below-ground, sea level rise or repeated storm surges combined with evaporation and concentration of salts in the coastal dune habitats, under climate change. Our approach complements experiments applying pulses of salinity stress to simulate extreme climatic events, which have documented considerable vulnerability of coastal plants to salinity (White *et al.*, 2014; Hanley *et al.*, 2017, 2019b).

MATERIALS AND METHODS

Study system

Jacquemontia sandwicensis, pā'ūohi'iaka in Hawaiian, is an endemic prostate vine that may span up to 3 m in diameter and lives ≥5 years (Wagner *et al.*, 1999). *Sida fallax*, 'ilima, is indigenous to Hawai'i and is widespread throughout the Pacific Islands to China. The coastal form of *S. fallax* is a prostrate subshrub with densely pubescent leaves, and is a short-lived perennial with a typical life span of ≥5 years (Wagner *et al.*, 1999). These species are abundant in the coastal strand communities and occur across all of the Hawaiian Islands (Warshauer *et al.*, 2009). They are comparable in habit, seed size (1.7–2.2 and 2–3 mm diameter in *S. fallax* and *J. sandwicensis*, respectively; Wagner *et al.*, 1999) and life span. Seeds used for the experiments were collected on O'ahu Island, at Ka'iwi State Scenic Shoreline and Ka'ena Point State Park, in June and July 2017. Seeds were collected from approx. 20 maternal plants at each site for both species (mean 3.79 mg and 3.45 mg seed mass for *J. sandwicensis* and *S. fallax*, respectively).

Experimental salinity treatments were developed using natural filtered seawater collected from the Ānuenuue Fisheries Research Center and diluted to the specified concentration with distilled water. Coastal substrate salinity levels are highly variable and can range from 1 ppt to 30 ppt NaCl (Barbour *et al.*, 1985) and, in tropical and sub-tropical regions, substrate salinity levels of 20–30 ppt have been reported (Johnson, 1977). The salinity treatments for these experiments span this range:

0, 3, 10 and 20 ppt NaCl. The lowest salinity treatment level (3 ppt) is approximately the level at which soils are considered saline, and crop yields are significantly reduced at 2.34 ppt (Munns and Tester, 2008). The highest treatment (20 ppt) exceeds the salinity levels required by many halophytes for optimal performance (Munns and Tester, 2008).

Germination experiment

Salinity tolerance in the earliest ontogenetic stage was tested with germination trials conducted in the lab, beginning in December 2017. Seeds were scarified with a razor to break physical dormancy and placed in a humidity chamber for 24 h to avoid cellular shock at first salinity treatment. Seeds were then placed in Petri dishes on germination blotter paper (Anchor Steel Blue Seed Germination Blotter), moistened with assigned salinity treatment solution (0, 3, 10 and 20 ppt NaCl) and sealed with parafilm to reduce evaporation loss. Eight replicate Petri dishes were prepared, each with 21–22 seeds, per treatment. This produced a total sample size of $n = 683$ seeds for *J. sandwicensis* and $n = 696$ seeds for *S. fallax* (4 salinity treatments $\times$ 8 replicate dishes $\times$ 21–22 seeds). Throughout the germination trials, additional solution was added as needed to maintain an even saturation of blotter paper, but also leading to increases in salinity over the duration of the experiment in similar ways between species. Petri dishes were placed on benches in the lab, and their positions were re-randomized weekly. Seeds were inspected daily and tallied as germinated when the radicle emerged at least 1 mm through the seed coat. Following cessation of germination for four consecutive days, ungerminated seeds from the three salinity solution treatments were rinsed with fresh water and transferred to Petri dishes with blotter paper moistened solely with distilled water to test germination rates following recovery from salinity exposure (Pujol *et al.*, 2000). Temperature was maintained between 21 and 22 °C, and humidity ranged from 60 to 65 % throughout the experiment.

Germination was analysed as the total percentage and with a modified Timmons Index (TI), which calculates germination velocity: $TI = \Sigma G/t$, in which G is the percentage of seed germination at 1 d intervals and t is the total germination period (Khan and Ungar, 1997). To analyse recovery germination following transfer of ungerminated seeds to distilled water, the recovery germination percentage (RGP) was calculated as $RGP = [a(c - b)] \times 100$, in which a is the total number of seeds germinated after transfer, b is the total number of seeds germinated under salinity treatment and c is the total seed count for each dish (Pujol *et al.*, 2000). To assess recovery germination, the RGP of the three salinity solution treatments was analysed with the 0 ppt group from the initial germination trial as control (Pujol *et al.*, 2000).

Greenhouse experiment

To investigate salinity tolerance across later ontogenetic stages, a pair of greenhouse experiments were conducted at the University of Hawai‘i at Mānoa (Honolulu, HI, USA) Pope

Greenhouse facility in which plants were exposed to the same range of salinity treatments. To test immediate responses in growth and physiological performance to salinity, the first experiment (hereafter referred to as the ‘short-term’ experiment) included trait analysis and biomass quantified by harvesting plants on the last day of the 3 week salinity treatments. To assess longer term effects of salinity on plant growth and reproduction, the second experiment (hereafter referred to as the ‘long-term’ experiment) allowed plants from all treatment groups to grow into maturity, at which time they were harvested for biomass data. Species were tested sequentially, from September 2017 to May 2018 for *J. sandwicensis* and from February to July 2018 for *S. fallax*.

Seeds were scarified and sown in flats filled with a 1:1:1 ratio by volume of Promix BX, black cinder and coral sand, a medium which approximates soil composition in the coastal dune habitats of these species. Seedlings were transplanted at the two-leaf stage into plastic pots filled with the same growth medium and treated with a single application of Osmocote slow-release fertilizer. At the time of transplant, plants were randomly assigned to treatment groups that differed with respect to three factors: salinity treatments (0, 3 and 20 ppt NaCl), ontogenetic stage (salinity imposed at the seedling, juvenile or mature stage) and experiment (short- or long-term harvest). After transplant, all plants were watered daily to saturation with tap water until the salinity treatments were initiated at the seedling (five leaves), juvenile (ten leaves) or mature (presence of buds or flowers) stage. Due to differences in growth rates and phenology, the chronological ages of these ontogenetic stages differed between *J. sandwicensis* and *S. fallax* (seedling, 2 weeks both species; juvenile, 5 vs. 8 weeks, respectively; mature, 24 vs. 16 weeks, respectively). At each of these ontogenetic stages, the salinity treatment was applied for 3 weeks. During these experimental periods, plants were watered twice a week with either tap water (control group), 3 ppt (low treatment) or 20 ppt (high treatment) salinity solutions. While this continuous application of salinity treatments over the experimental period leads to increases in salinity via salt accumulation over time, this reflects natural repeated storm surges or coastal flooding events, which similarly would increase salinity in coastal habitats, particularly those with high solar radiation and thus high drying capacity such as Hawaiian coastal dune sites (Price *et al.*, 2012; Geng *et al.*, 2016).

Plants in the short-term experiment were subjected to leaf trait analysis on the final day of the salinity treatments in order to assess plastic responses to salinity treatments and were harvested to assess short-term effects of salinity on growth. The youngest mature leaf (fully expanded and lignified) was used for all trait analyses. Stomatal conductance ($\text{mmol m}^{-2} \text{s}^{-1}$) was measured with an SC-1 steady-state leaf porometer (Decagon Devices, Pullman, WA, USA). A chlorophyll meter (SPAD-502 Plus, Konica Minolta Inc., Japan) was used to measure chlorophyll content (index scale –9.9 to 199.9). Net photosynthesis ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{s}^{-1}$), transpiration ($\text{mmol H}_2\text{O m}^{-2} \text{s}^{-1}$) and instantaneous WUE (net photosynthesis/transpiration) were measured using a Li-Cor 6400 (Li-Cor Inc., Lincoln, NE, USA). For all physiological measurements, photosynthetically active radiation was set to $500 \mu\text{mol m}^{-2} \text{s}^{-1}$, ambient CO_2 concentration was set to $400 \mu\text{mol m}^{-2} \text{s}^{-1}$, mean leaf temperature

was 30 °C (26–35 °C) and mean relative humidity was 57 % (49–69 %). Three morphological traits were measured on the same leaf used for measuring physiological traits. Leaf thickness (mm) was measured with a micrometer. Leaf area (cm²) was computed with ImageJ v. 1.51j from a digital photo of the leaf (US National Institutes of Health, Bethesda, MD, USA). The leaf was then oven-dried at 60 °C and weighed using an analytical balance to obtain the LMA (g mm⁻²), calculated by dividing the oven-dried mass by the leaf area. Plants were then measured for height (cm), i.e. length of the longest stem, and were harvested in order to obtain total biomass and biomass allocation data. The total sample size for the short-term experiment was $n = 108$ (3 salinity treatments $\times$ 3 ontogenetic stages $\times$ 12 replicate plants).

Due to pest infestations during the *S. fallax* experiments, only *J. sandwicensis* was included in the long-term experiment. In the long-term experiment, plants in all treatment groups were harvested 2 weeks after the end of the mature stage treatment period, at 29 weeks. This design means that plants exposed to salinity treatments in early ontogenetic stages had a longer recovery period before being harvested than later ontogenetic stages. The long-term experiment includes the low (3 ppt) and high (20 ppt) salinity treatments applied at each ontogenetic stage (seedling, juvenile and mature), resulting in six salinity treatment groups, as well as a single control group that received tap water throughout the duration of the 29 week experimental period, giving a total sample size of $n = 84$ (2 salinity treatments $\times$ 3 ontogenetic stages $\times$ 12 replicate plants + 12 control plants).

For all plants in the short- and long-term experiments, roots and shoots were harvested separately, oven-dried at 60 °C and weighed using an analytical balance. The root/shoot ratio was calculated from oven-dried mass values. The onset of flowering was recorded throughout the experiment to test if salinity treatments led to shifts in phenology.

Statistical analyses

All germination metrics were analysed with analysis of variance (ANOVA) models with salinity treatment as a fixed factor using Petri dish as the unit of replication. When salinity effects were significant, pair-wise differences among salinity levels

were assessed with Turkey's HSD (honestly significant difference) tests. *Sida fallax* RGP values did not meet normality or homogeneity of variance assumptions for ANOVA and were analysed with Kruskal-Wallis and Dunn's tests to determine significant differences among groups. All analyses were done in R v. 3.5.0 with base (R Core Team, 2018), dplyr (Wickham *et al.*, 2018), car (Fox *et al.*, 2011), multcomp (Hothorn *et al.*, 2008), dunn.test (Dinno, 2017) and the germinationmetrics package (Aravind *et al.*, 2018).

Tolerance was analysed as the effect of salinity treatments on survival, flowering and total biomass in the short- and long-term experiments, revealing immediate and long-term effects of salinity on performance. Survivorship in the greenhouse experiment was analysed with binomial generalized linear models (GLMs). Binomial GLMs were used to determine if salinity influenced the likelihood that plants flowered during the experiment with salinity and ontogeny as predictors. To investigate whether salinity shifts the timing of reproduction, GLMs were used to analyse onset of flowering as the response to salinity and ontogenetic stage. Since the mature stage was defined as the presence of buds or flowers, only plants exposed to salinity at the seedling and juvenile stages were included in this analysis. ANOVA was used to assess the effects of salinity and ontogeny, as well as their interaction, on total biomass. When main effects and interactions were significant, pair-wise differences were assessed with Tukey's HSD tests. Comparable biomass in control vs. salinity-treated plants would indicate complete salinity tolerance, while lower biomass in salinity-treated compared with control plants would reveal incomplete tolerance.

Trait variation measured in the short-term group was assessed using ANOVA models to test the effects of salinity, ontogeny and their interaction. Significant effects of salinity reveal phenotypic plasticity in traits in response to salinity, and significant effects of ontogeny indicate ontogenetic shifts in trait expression. Significant interactions between salinity and ontogeny indicate that plasticity to salinity shifts across plant ontogeny. Significant main effects and interactions were assessed with Tukey's HSD test for insights into pair-wise differences.

To identify traits associated with salinity tolerance across ontogeny, additional models testing the effects of salinity and ontogeny on total biomass were run, with traits included as covariates (ANCOVA). In these models, significant trait effects reveal that those traits are linked with biomass (i.e. that

TABLE 1. Seed performance metrics including germination percentage, modified Timmons Index (mTI) and recovery germination percent (RGP) measured for control and salinity treatment groups

Salinity (ppt NaCl)	Germination (%)	Rate (mTI)	RGP (%)
<i>J. sandwicensis</i>			
0	86.04 $\pm$ 1.73 ^a	72.92 $\pm$ 1.44 ^a	—
3	83.10 $\pm$ 3.88 ^a	70.21 $\pm$ 3.13 ^{ab}	55.65 $\pm$ 19.47 ^{ab}
10	76.61 $\pm$ 4.16 ^a	61.37 $\pm$ 3.18 ^b	38.66 $\pm$ 8.55 ^b
20	47.15 $\pm$ 3.78 ^b	32.45 $\pm$ 3.00 ^c	49.75 $\pm$ 6.32 ^b
<i>S. fallax</i>			
0	74.23 $\pm$ 1.92 ^a	90.8 $\pm$ 2.26 ^a	—
3	59.59 $\pm$ 2.53 ^a	82.18 $\pm$ 3.19 ^b	58.57 $\pm$ 12.47 ^b
10	9.82 $\pm$ 1.49 ^b	17.82 $\pm$ 2.37 ^c	76.29 $\pm$ 2.40 ^{ab}
20	—	—	66.39 $\pm$ 2.93 ^b

Mean values $\pm$ 1 s.e. ($n = 21$ – 22) are reported. Tukey HSD letters show significant variation among treatments. Results not shown (—) in groups with effectively no germination (0–1 seed germinating).

they are functional). Significant interactions between traits and salinity indicate that trait functionality is contingent on salinity. Specific mechanisms of tolerance could be identified when traits are related to biomass under high salinity, but not under control conditions. Significant interactions between traits, salinity and ontogeny indicate that trait functionality is contingent on salinity and ontogeny. If the three-way interaction was not significant, it was removed from the model. Slopes of significant interactions were tested with Tukey's HSD test to assess the effect of each salinity treatment group on biomass.

Where assumptions of normality and homogeneity of variance were not met, data were transformed and, if necessary, influential outliers based on Cook's Distance were removed. All analyses were done in R v. 3.5.0 with base (R Core Team, 2018), dplyr (Wickham *et al.*, 2018), car (Fox *et al.*, 2011), multcomp (Hothorn *et al.*, 2008), dunn.test (Dinno, 2017) and lsmeans (Lenth, 2016).

RESULTS

Germination experiment

Salinity treatments significantly influenced germination in both species, with the lowest percentages and rates at the highest salinity level, indicating that these species are not fully tolerant to high salinity in the earliest ontogenetic stage (Table 1). For *J. sandwicensis*, salinity significantly reduced both the germination rate (TI, $F_{3,28} = 44.30$, $P < 0.0001$) and percentage germination ($F_{3,28} = 25.64$, $P < 0.0001$, Table 1). Previous salinity

treatment significantly reduced recovery germination percentage ($F_{3,24} = 7.13$, $P = 0.001$, Table 1), revealing that salinity exposure may have damaged seeds.

Germination was lower overall in *S. fallax* than in *J. sandwicensis* (Table 1) and showed high sensitivity to salinity for both percentage germination ($F_{2,21} = 221.83$, $P < 0.0001$) and germination rate (TI, $F_{2,21} = 277.39$, $P < 0.0001$), with only a single seed germinating in the 20 ppt high salinity group. Previous salinity treatment at 3 and 20 ppt significantly reduced recovery percentage germination ($\chi^2 = 11.43$, d.f. = 3, $P < 0.01$), but not at the 10 ppt treatment level, indicating that for *S. fallax*, even though high salinity inhibited germination, moderate salinity levels probably did not damage seeds, allowing for them to germinate upon rinsing with fresh water.

Short- and long-term experiments: plant survival and reproduction

Survivorship was high for both species across all salinity groups in both the short- and long-term experiments: 96.4 % in *J. sandwicensis* and 93.5 % in *S. fallax*. For *J. sandwicensis*, survivorship was significantly influenced by salinity ($\chi^2 = 7.37$, d.f. = 2, $P < 0.001$), with most mortalities occurring under high salinity (20 ppt) at seedling and juvenile stages. For *S. fallax*, survivorship was significantly influenced by salinity ($\chi^2 = 6.42$, d.f. = 2, $P < 0.001$), and this effect was contingent on ontogeny (significant interaction between salinity and ontogeny: $\chi^2 = 6.12$, d.f. = 2, $P < 0.05$), with most mortalities occurring under high salinity (20 ppt) at the seedling stage.

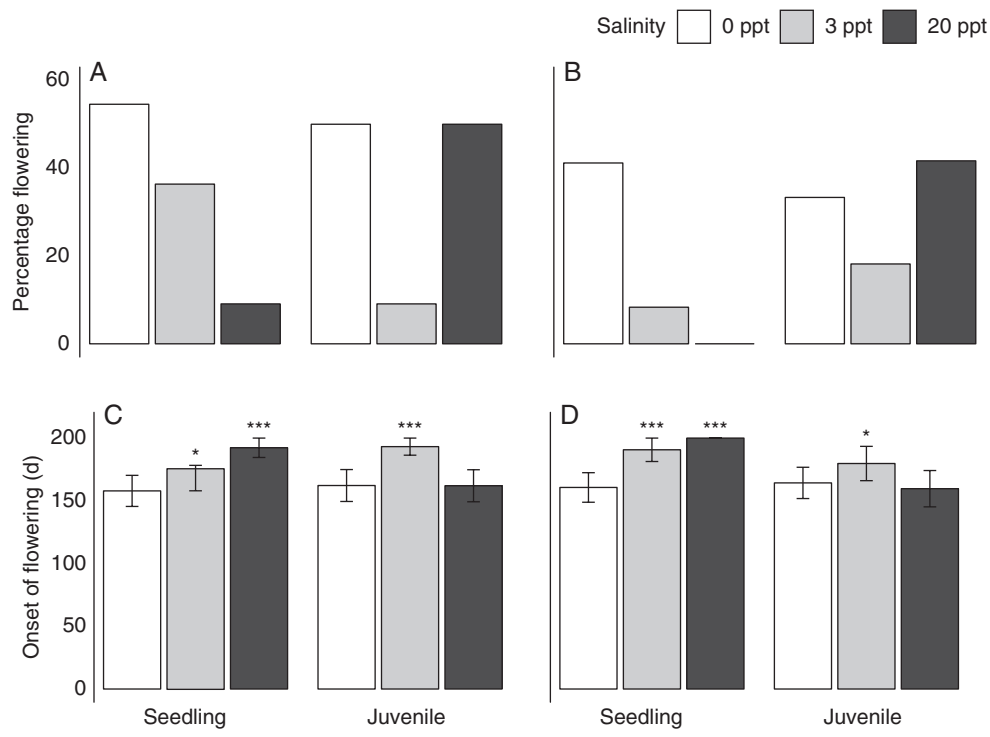


FIG. 1. Percentage of individuals that reached reproductive maturity in (A) *J. sandwicensis* and (B) *S. fallax*, and effect of salinity on the onset of flowering in (C) *J. sandwicensis* and (D) *S. fallax* after the recovery period following a 3 week salinity treatment imposed at the assigned ontogenetic stage. Bars are means $\pm$ 1 s.e. ($n = 10-12$). Significant differences among salinity groups within ontogenetic stages is shown as: *** $P < 0.0001$; ** $P < 0.01$; * $P < 0.05$.

TABLE 2. ANOVA results for the effects of ontogeny (seedling, juvenile and mature), salinity (0, 3 and 20 ppt NaCl) and their interaction on plant performance after a 3 week salinity treatment imposed at the assigned ontogenetic stage (short-term experiment)

Variable	d.f.	Ontogeny (F)	Salinity (F)	O × S (F)
<i>J. sandwicensis</i>				
Dry mass (g)	93	127.85***	1.18	1.33
Height (cm)	61	368.58***	1.93	0.65
Root/shoot	89	8.15***	0.81	2.85*
<i>S. fallax</i>				
Dry mass (g)	97	401.70***	3.46*	4.63**
Height (cm)	59	244.07***	1.65	4.63*
Root/shoot	95	101.05***	7.72***	4.17 **

All factors were treated as fixed (F). Significance is given as: *** $P < 0.0001$; ** $P < 0.01$; * $P < 0.05$.

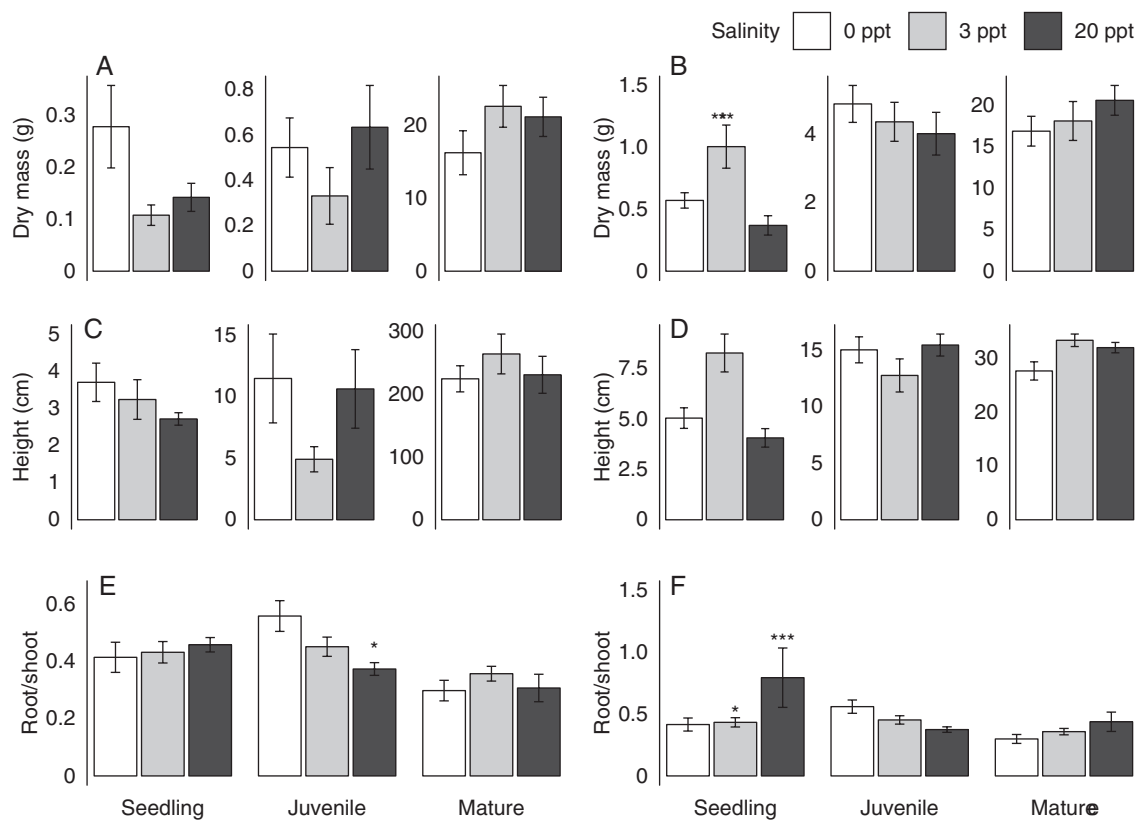


FIG. 2. Effect of salinity on growth and biomass allocation from the short-term experiment, in which plants were harvested on the last day of the 3 week salinity treatments imposed at the assigned ontogenetic stage: *J. sandwicensis* (A) total dry biomass, (B) height estimated as the length of the longest stem and (C) root/shoot ratio; and *S. fallax* (D) total dry biomass, (E) height estimated as the length of the longest stem and (F) root/shoot ratio. Bars are means $\pm$ 1 s.e. (n = 10–12). Significant differences among salinity groups within ontogenetic stages are shown as: *** $P < 0.0001$; ** $P < 0.01$; * $P < 0.05$.

For both species, salinity exposure altered reproductive phenology (Fig. 1). For *J. sandwicensis*, there was no overall effect of salinity on the likelihood of flowering ($\chi^2 = 5.73$, d.f. = 2, $P = 0.06$), but this depended on ontogenetic stage (significant interaction between salinity and ontogeny: $\chi^2 = 7.02$, d.f. = 2, $P = 0.03$). Moreover, for plants that reproduced, salinity significantly delayed flowering ($\chi^2 = 40.06$, d.f. = 2, $P < 0.0001$), and this effect was also contingent on ontogeny ($\chi^2 = 36.10$, d.f. = 2, $P < 0.0001$). Specifically, seedlings exposed to salinity were less likely to flower at all and flowered significantly later than control

plants, while juvenile stage plants delayed flowering only under moderate salinity (Fig. 1).

For *S. fallax*, salinity reduced the likelihood of flowering ($\chi^2 = 10.55$, d.f. = 2, $P < 0.001$), and this effect was contingent on ontogeny ($\chi^2 = 7.44$, d.f. = 2, $P = 0.02$). Salinity also delayed the onset of flowering ($\chi^2 = 46.93$, d.f. = 2, $P < 0.0001$), and depended on ontogeny ($\chi^2 = 37.88$, d.f. = 2, $P < 0.0001$). Similar to *J. sandwicensis*, salinity exposure early in ontogeny had the strongest effects on reproductive phenology, with juvenile stage plants demonstrating relatively high tolerance to high salinity (Fig. 1).

TABLE 3. ANOVA results for the effects of ontogeny (seedling, juvenile and mature), salinity (0, 3 and 20 ppt NaCl) and their interaction (all fixed factors) on physiological and morphological traits after a 3 week salinity treatment imposed at the assigned ontogenetic stage (short-term experiment)

Response variable	d.f.	Ontogeny (F)	Salinity (F)	OxS (F)
<i>J. sandwicensis</i>				
Photosynthesis ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$)	61	28.12***	2.33	5.57***
Transpiration ($\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$)	61	16.17***	26.51 ***	5.28**
WUE ($\mu\text{mol CO}_2 \text{ mol}^{-1} \text{ H}_2\text{O}$)	59	16.42***	22.18***	18.57***
Conductance ($\text{mmol m}^{-2} \text{ s}^{-1}$)	61	33.02***	31.53***	2.31
Chlorophyll index	61	2.60	5.47 **	2.32
LMA (g mm^{-2})	60	19.10***	18.67***	2.86*
Leaf thickness (mm)	58	55.38***	24.21***	22.93***
<i>S. fallax</i>				
Photosynthesis ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$)	61	22.69***	32.22***	10.38***
Transpiration ($\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$)	56	42.28***	39.97***	9.09***
WUE ($\mu\text{mol CO}_2 \text{ mol}^{-1} \text{ H}_2\text{O}$)	59	13.18***	1.86	5.38***
Conductance ($\text{mmol m}^{-2} \text{ s}^{-1}$)	58	6.96**	21.03***	2.28
Chlorophyll index	61	2.04	1.65	2.98*
LMA (g mm^{-2})	61	182.27***	6.84**	0.45
Leaf thickness (mm)	61	9.81 ***	9.51***	2.15

Significance is given as: *** $P < 0.0001$; ** $P < 0.01$; * $P < 0.05$.

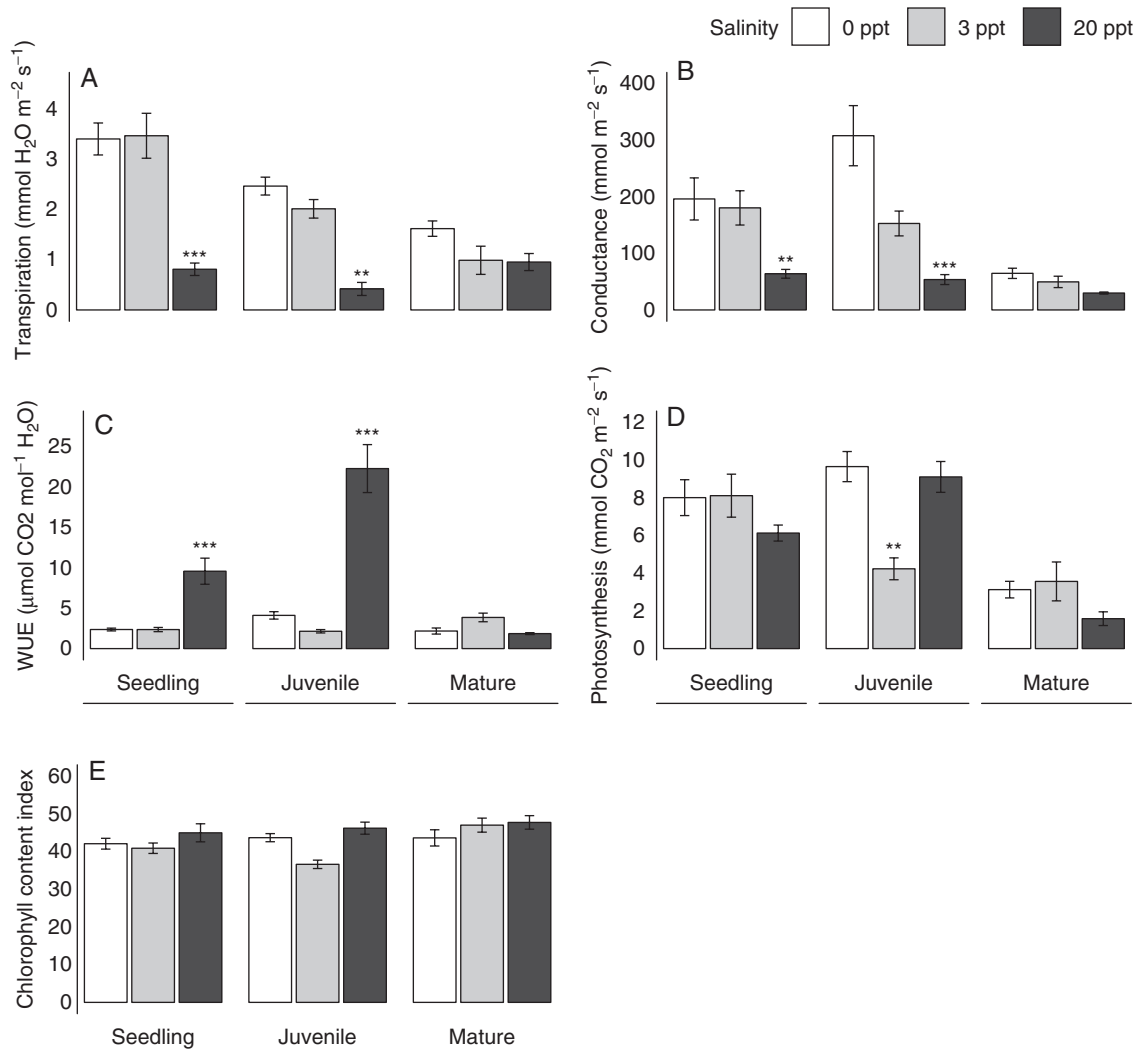


FIG. 3. Effect of salinity on *J. sandwicensis* physiological traits from the short-term experiment, in which traits were measured on the last day of the 3 week salinity treatments imposed at the assigned ontogenetic stage: (A) transpiration, (B) stomatal conductance, (C) water use efficiency, (D) photosynthesis and (E) chlorophyll content. Bars are means ± 1 s.e. (n = 10–12). Significant differences among salinity groups within ontogenetic stages are shown as: *** $P < 0.0001$; ** $P < 0.01$; * $P < 0.05$.

Short-term experiment: growth

When quantified at the end of the 3 week salinity treatments, both species demonstrated maintained growth in terms of total biomass and height (Table 2; Fig. 2). Elevated growth was even detected for *S. fallax* seedlings under low salinity, indicating high tolerance to low salinity. Contrary to our prediction that salinity-induced osmotic stress would lead to an increase in the root/shoot ratio, we observed a significant decrease in the root/shoot ratio in the juvenile stage of *J. sandwicensis* (Tukey $P < 0.05$) and at the seedling stage in *S. fallax* (Tukey $P < 0.02$).

Short-term experiment: trait plasticity

Considerable trait variation was detected in both species, indicating that trait expression is highly plastic in response

to salinity, that expression changes across plant ontogeny and that there are ontogenetic shifts in plasticity of the traits (Table 3). Both species experienced declines in transpiration and stomatal conductance rates under high salinity early in ontogeny (Table 3; Figs 3 and 4). In *J. sandwicensis*, photosynthesis was generally consistent across stages and salinity treatments, leading to significantly higher WUE for seedlings (Tukey $P < 0.001$) and juvenile plants (Tukey $P < 0.001$) in the 20 ppt salinity groups (Fig. 3C). In contrast, photosynthetic rates declined in conjunction with conductance and transpiration in *S. fallax* salinity groups, with no significant effects on WUE early in ontogeny (Table 3; Fig. 4). Chlorophyll content was consistent across ontogeny and under salinity exposure in both species, with the exception of an increase under high salinity in *S. fallax* juvenile plants (Table 3; Figs 3 and 4).

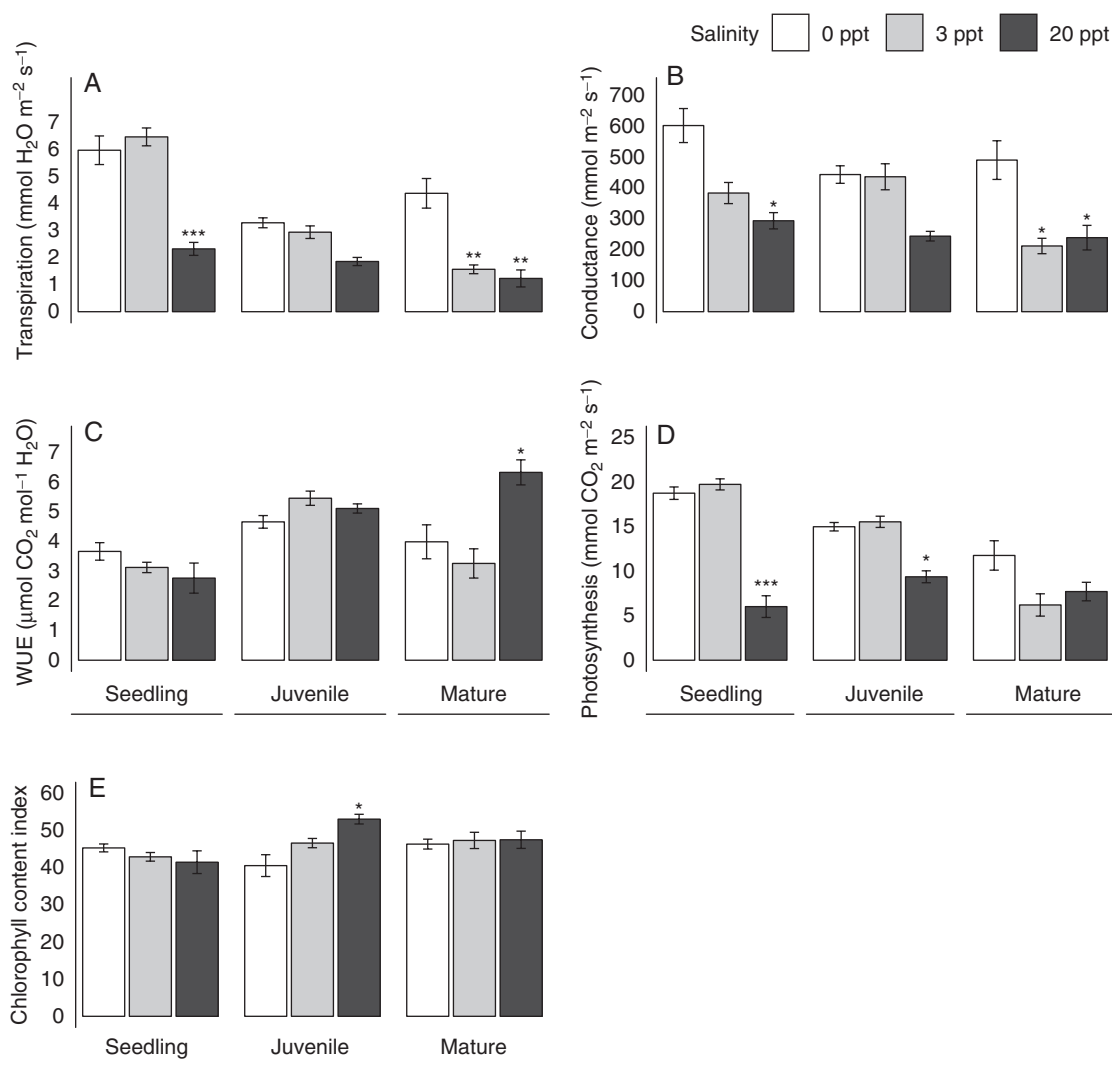


FIG. 4. Effect of salinity on *S. fallax* physiological traits from the short-term experiment, in which traits were measured on the last day of the 3 week salinity treatments imposed at the assigned ontogenetic stage: (A) transpiration, (B) stomatal conductance, (C) water use efficiency, (D) photosynthesis and (E) chlorophyll content. Bars are means ± 1 s.e. (n = 10–12). Significant differences among salinity groups within ontogenetic stages are shown as: *** $P < 0.0001$; ** $P < 0.01$; * $P < 0.05$.

All gas exchange parameters generally declined from the seedling to juvenile and mature stages (Figs 3 and 4). In *J. sandwicensis*, no traits were plastic in response to salinity at the mature stage (revealing a decline in plasticity across ontogeny), indicating a strong role of ontogeny in the ecophysiology of this species. In contrast, for *S. fallax*, reductions in gas exchange parameters under high salinity were observed at all ontogenetic stages (Fig. 4).

Leaf morphology was also plastic in response to salinity, particularly early in ontogeny (Table 3; Fig. 5). In both species, seedlings under high salinity developed significantly thicker leaves (Fig. 5A, B). High salinity also led to greater LMA in *J. sandwicensis* seedling and juvenile plants (Table 3; Fig. 5C, D).

Short-term experiment: trait links to growth

The ANCOVA models revealed traits linked with plant growth under salinity (see Supplementary data Table S1 for full results). Both species demonstrated a significant role for leaf thickness in salinity tolerance, although the patterns differed between species (Fig. 6). In *J. sandwicensis*, a significant leaf thickness $\times$ salinity interaction ($F_{2,55} = 4.26$, $P = 0.02$) revealed that plants with thinner leaves tended to have greater biomass across all salinity groups, although this relationship was strongest for plants in the 3 ppt NaCl treatment, revealing a benefit to thin leaves under low salinity conditions (Fig. 6A). In contrast, for *S. fallax*, the significant leaf thickness $\times$ salinity interaction ($F_{2,56} = 6.03$, $P < 0.01$) reflected a benefit in biomass

for plants with thicker leaves, although this relationship was partially decoupled under high salinity (Fig. 6B).

For *J. sandwicensis*, three-way interactions were significant for WUE ($F_{4,47} = 6.61$, $P < 0.0001$) and chlorophyll content ($F_{4,54} = 4.26$, $P < 0.05$). There was a positive link between WUE and biomass under control conditions early in ontogeny and under high salinity in the juvenile stage (Fig. 7A). However, a negative relationship between WUE and biomass was observed under high salinity in the seedling and mature stages (Fig. 7A), indicating that salinity seems to compromise WUE functionality at those stages. Plants with greater chlorophyll content tended to be larger in all salinity treatments across all ontogenetic stages, although the magnitude of this relationship was greater under high salinity in juvenile plants (Fig. 7B), revealing chlorophyll content to be an important salinity tolerance trait at the juvenile stage.

For *S. fallax*, a significant three-way interaction was only detected for transpiration rate ($F_{4,52} = 2.79$, $P = 0.03$). Greater transpiration rates are linked with larger biomass under moderate salinity in seedlings and juvenile plants, while plants with more conservative transpiration (i.e. lower rates) have greater biomass under high salinity in seedling and mature stages (Fig. 7C). These patterns indicate that plants can modulate gas exchange to optimize growth in variable ways across ontogeny.

Long-term experiment: growth

For *J. sandwicensis*, final total biomass measured at 29 weeks was significantly influenced by salinity treatments ($F_{2,67} = 8.23$,

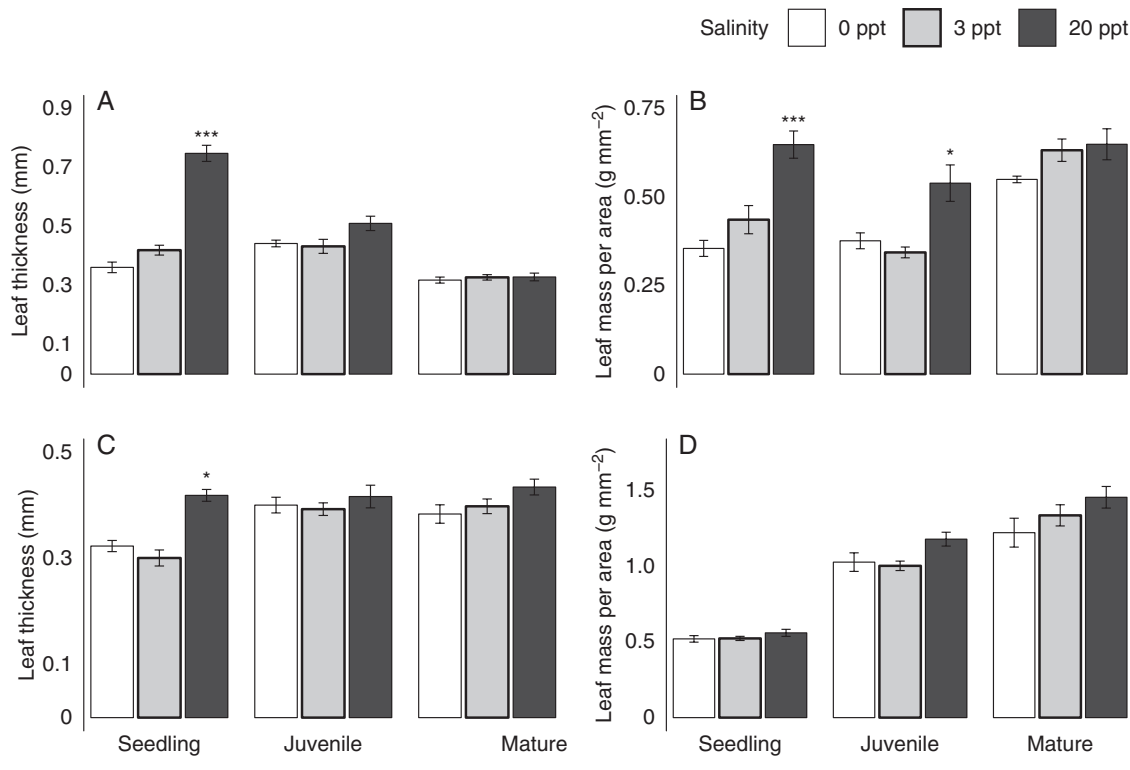


FIG. 5. Effect of salinity on morphological traits in *J. sandwicensis* (A, B) and *S. fallax* (C, D) from the short-term experiment, in which traits were measured on the last day of the 3 week salinity treatments imposed at the assigned ontogenetic stage: (A, C) leaf thickness and (B, D) leaf mass per area. Bars are means ± 1 s.e. ($n = 10$ –12). Significant differences among salinity groups within ontogenetic stages are shown as: *** $P < 0.0001$; ** $P < 0.01$; * $P < 0.05$.

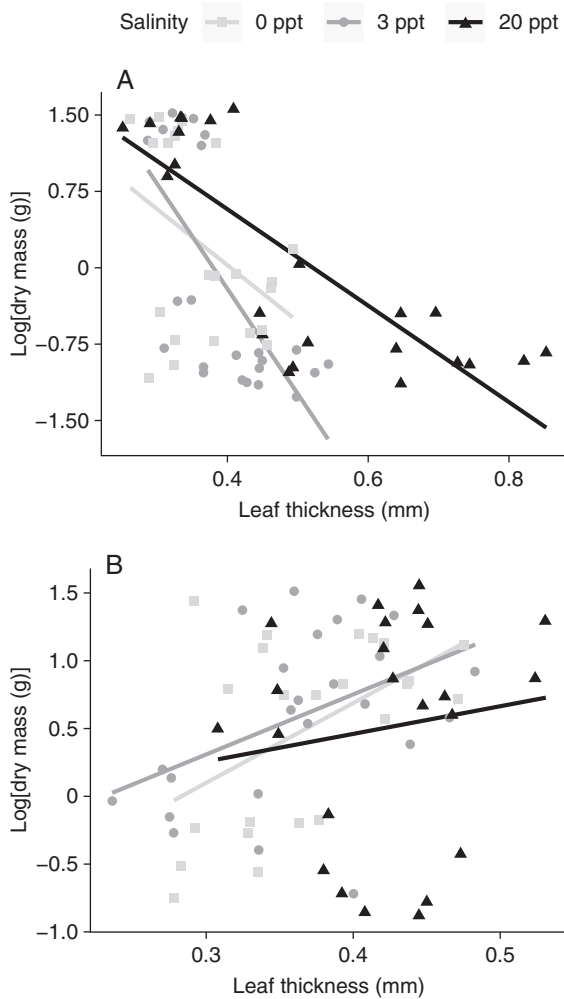


FIG. 6. Relationships between leaf thickness and biomass differ across salinity treatments as revealed by trait $\times$ salinity interactions in ANCOVA in (A) *J. sandwicensis* and (B) *S. fallax*. Plants were harvested and traits were measured on the last day of the 3 week salinity treatments imposed at the assigned ontogenetic stage.

$P < 0.001$). While plants were fully tolerant to low salinity treatment at all ontogenetic stages, exposure to high salinity at the seedling stage resulted in a significant reduction in biomass (Fig. 8). This indicates that salinity tolerance is most limited at the seedling stage and to high salinity levels.

DISCUSSION

Globally, little is known about the salinity tolerance of native coastal plant species and how tolerance may vary across ontogeny. Due to rising sea levels and increases in extreme storms and coastal flooding, coastal habitats are increasingly threatened by high salinity, and identification of salinity thresholds and modes of tolerance are urgently needed to conserve coastal plants (Hanley et al., 2017, 2019b). We found that salinity tolerance, quantified on the basis of survival, growth and reproduction, varied considerably between two coastal dune species

from Hawai'i, and that tolerance shifted across plant ontogeny in complex ways.

Overall, *J. sandwicensis* salinity tolerance increased across ontogeny, with seeds and seedlings being highly sensitive, juveniles moderately tolerant and mature plants unaffected by salinity exposure. For *S. fallax*, seeds demonstrated some tolerance to salinity with recovery germination percentages and, although high salinity suppressed physiological performance in seedlings, moderate salinity actually enhanced seedling growth. Salinity tolerance was relatively high in juvenile and mature *S. fallax* stages. Considering the multiple fitness metrics, we observed that by some metrics, *S. fallax* appears to be more sensitive to salinity exposure than *J. sandwicensis*, as evidenced by longer delays in flowering and decreases in physiological performance under salinity at all ontogenetic stages. However, when considering growth (biomass and height), *S. fallax* appears to be more tolerant to salinity than *J. sandwicensis*, and to achieve this tolerance earlier in ontogeny. These patterns highlight the value of considering multiple fitness metrics as some performance variables are more sensitive to salinity than others, and these responses can differ across ontogeny.

Germination percentage and rate were significantly reduced under high salinity for both species, although *S. fallax* in particular demonstrated relatively high recovery germination following rinsing with fresh water. Germination typically occurs during the wetter winter months when ample rainfall reduces salinity in the uppermost layer of soil (Ungar, 1982), and reduced rates of germination under salinity may reflect an adaptive strategy of seeds to withstand unfavourable conditions until they can germinate under more favourable conditions (Khan and Gul, 2006). However, under climate change, the decoupling of germination cues and the most saline and drier post-germination environment will exert strong selection on seedling traits (Donohue et al., 2010) if coastal dune plants are to persist.

Low seedling salinity tolerance is a well-documented pattern in other species (Alpha et al., 1996; Houle et al., 2001; Vicente et al., 2004; Lauchli and Grattan, 2011), and is likely to be due to the negative impacts of both osmotic stress and ion toxicity (Cordeiro et al., 2014). Seedlings have underdeveloped root systems and reduced water absorption capacity (Cordeiro et al., 2014), making them particularly vulnerable to the osmotic stress induced by salinity exposure. Even into the juvenile stage, developing plants are still increasing carbon capture (Mason et al., 2013), which may limit stored reserves and general potential to tolerate stress. Consistent with these patterns, *J. sandwicensis* demonstrated low tolerance to salinity at the seedling stage in terms of both short- and long-term growth. However, juvenile *J. sandwicensis* plants appear to fully tolerate even high salinity, indicating that for this species, salinity tolerance increases from the seedling to juvenile stage. Early onset of salinity tolerance was observed in *S. fallax* with an increase in biomass for seedlings exposed to low (3 ppt) salinity, although this pattern dissipated by the juvenile stage. Salt-stimulated growth at the seedling stage has not been previously reported, to our knowledge, even among known salt-tolerant species (Ungar, 1991; Flowers et al., 2010). Perhaps this response in *S. fallax* is a consequence of its acquisitive resource strategy, as high rates of

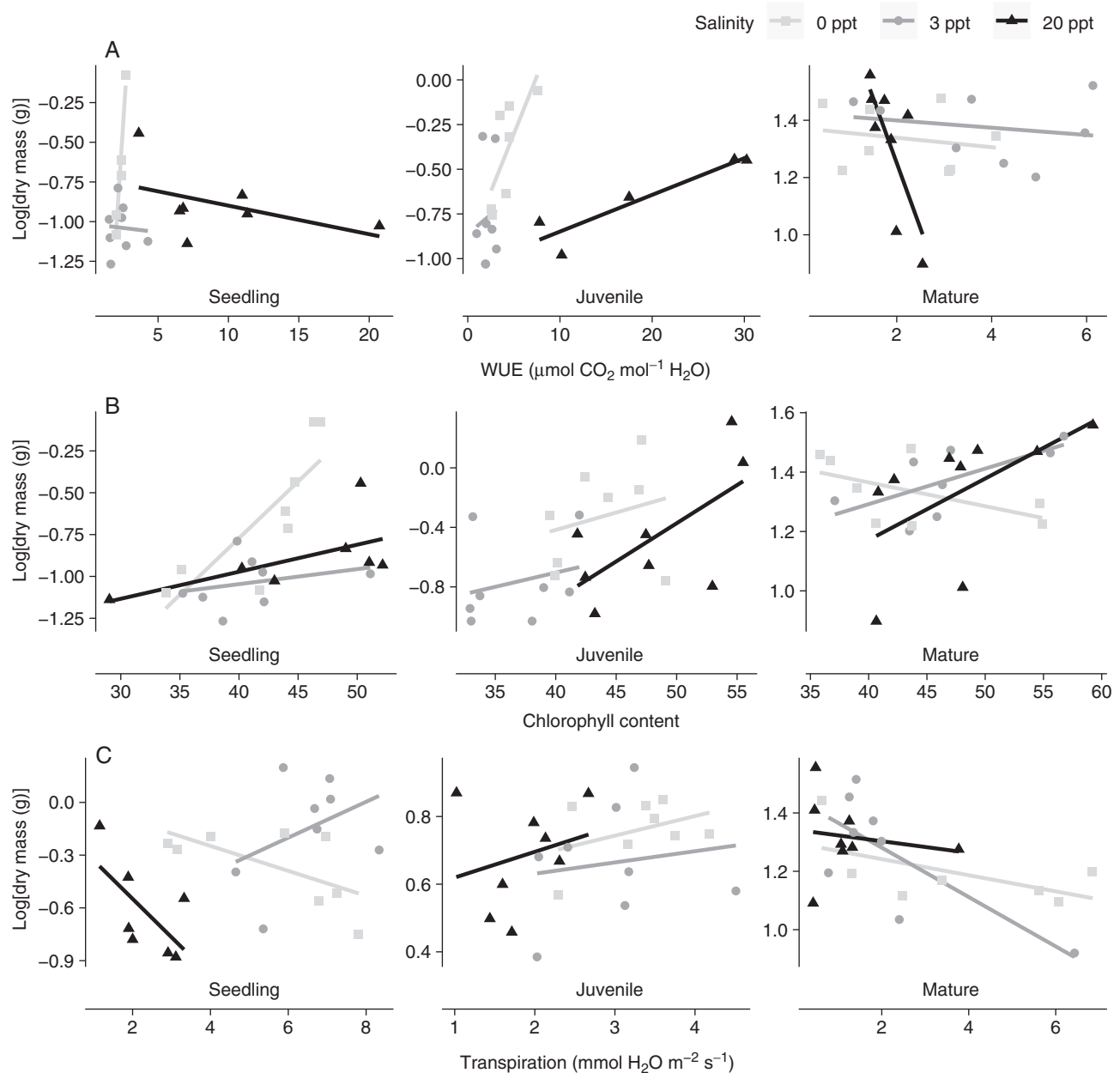


FIG. 7. Relationships between traits and biomass vary across salinity treatments in different ways across ontogeny as revealed by trait $\times$ salinity $\times$ ontogeny interactions in ANCOVA in *J. sandwicensis*: (A) water use efficiency, (B) chlorophyll content; and *S. fallax*: (C) transpiration. Plants were harvested and traits were measured on the last day of the 3 week salinity treatments imposed at the assigned ontogenetic stage.

carbon fixation may outpace the toxic effects of salt ions at low levels (Munns, 2002; Munns and Tester, 2008).

For both species, salinity exposure at early ontogenetic stages led to significant delays in flowering, similar to the phenological delays observed in other systems (Van Zandt *et al.*, 2003; Boscaiu *et al.*, 2005; Kazan and Lyons, 2016). In this study, flowering delays were considerable, from 19 to 35 d for *J. sandwicensis* and from 30 to 40 d for *S. fallax*. Other research has shown no delay in onset of flowering for *S. fallax* under drought (Yorkston and Daehler, 2006), indicating that observed delays are probably salt-specific effects. Phenology is influenced by a variety of abiotic factors, such as photoperiod, temperature and precipitation

(Franks *et al.*, 2007), and it is unclear how variation in abiotic factors due to climate change will act synergistically with salinity-induced delayed flowering. Salinity-induced phenological shifts could cascade to affect species interactions, such as competition, pollination and herbivory (Forrest and Miller-Rushing, 2010). Moreover, delays and reductions in flowering may further diminish dispersal ability in fragmented coastal habitats as sea levels rise (Franks *et al.*, 2007).

To better understand why some species and stages are more vulnerable than others to high salinity, we examined a suite of leaf functional traits to characterize links between traits and growth under salinity. In general, the traits were highly plastic

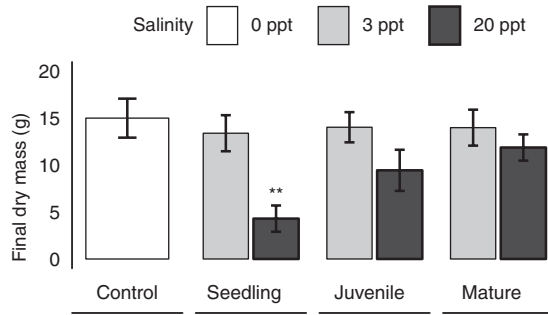


FIG. 8. Effect of salinity on total dry biomass of 29-week-old *J. sandwicensis* plants after a recovery period following the 3 week salinity treatment imposed at the assigned ontogenetic stage (long-term experiment). Bars are means $\pm$ 1 s.e. ($n = 10$ – 12). Significant differences among salinity groups within ontogenetic stages are shown as: *** $P < 0.0001$; ** $P < 0.01$; * $P < 0.05$.

in response to salinity, and shifted dramatically in expression across ontogeny. Moreover, the strength and direction of the trait–performance relationships varied across ontogeny and were contingent on salinity exposure. Because the trait relationships and trajectories were not similar between the two species, it will be difficult to generalize across species in coastal communities, even for species with similar habits in the same environments. Future research examining additional coastal species is needed before general models of salinity tolerance can be developed.

The considerable plasticity observed in trait response to salinity across ontogeny is consistent with our predictions. Coastal plant communities are exposed to strong abiotic stressors and variable conditions (Gagné *et al.*, 1999; Richmond and Mueller-Dombois, 1972), probably having resulted in strong selection pressure for phenotypic plasticity (Jump and Peñuelas, 2005) and adaptations to osmotic stress (Lloret *et al.*, 2004). High WUE, decreased stomatal conductance and increased LMA at the seedling and juvenile stages in response to salinity exposure indicate that *J. sandwicensis* is highly plastic, and in directions consistent with high osmotic tolerance (Munns, 2002; Munns and Tester, 2008). *Sida fallax* appears to be more salt sensitive at the ecophysiological trait scale, and was not able to maintain photosynthesis or transpiration under high salinity treatments at the seedling and juvenile stages, resulting in reduced biomass. While stomatal regulation can enhance survival during periods of stress by limiting water loss, in the long term, the resulting decreases in growth can constrain plant competitive ability, reproduction and overall fitness. Several studies have found reductions in photosynthesis under increasing salinity exposure (Goldstein *et al.*, 1996; Santiago *et al.*, 2000; Naumann *et al.*, 2007; Redondo-Gómez *et al.*, 2007), probably due to stomatal closure to minimize water loss via transpiration, which in turn impacts photosynthetic capacity (Parida and Das, 2005), but the consequences for long-term fitness remain unclear. *Sida fallax* is highly plastic at all life stages, which may be beneficial in a rapidly changing climate (Nicotra *et al.*, 2010), but, on the other hand, the conservative leaf economic strategy of *J. sandwicensis* may be advantageous under predicted climate change stressors, such as extended periods of drought (Chapin *et al.*, 1993). Whether an acquisitive and plastic growth strategy or a more conservative stress-tolerant strategy is more advantageous for coastal plants facing increases in salinity and drought stress under climate change requires further investigations linking these patterns to population dynamics across more species.

CONCLUSION

Coastal ecosystems are globally threatened and largely understudied. Due to the complexity of how climate change effects interact with plant size and ontogeny, it is difficult to predict plant vulnerabilities from studies examining stress tolerance at only a single stage. Most importantly, we found that these common coastal plants are not fully salt tolerant despite their abundance in these highly saline coastal sand dunes, and that salinity tolerance tended to increase across ontogeny. Sensitivity to salinity at early critical ontogenetic stages may have severe consequences as sea levels rise and precipitation regimes change, potentially creating a bottleneck to recruitment (Walck *et al.*, 2011) as stressors are amplified (Greaver and Sternberg, 2010). In fragmented habitats such as coastal areas, climate change may overwhelm the ability of low-density populations such as these to adapt and migrate under climate change (Jump and Peñuelas, 2005). Given the variation in tolerance detected between two species of similar habit, longevity and distribution, assessment of salinity tolerance of additional species is needed before generalizations can be made. Moreover, experiments testing different intensities and durations of salinity stress and the simultaneous effects of salinity and other co-occurring stressors (e.g. drought) will provide much needed insights into the likely resilience of coastal plants under climate change, enhancing conservation of these critical ecosystems.

SUPPLEMENTARY DATA

Supplementary data are available online at <https://academic.oup.com/aob> and consist of Table S1: summary of ANCOVA results for traits associated with tolerance, with total biomass as performance metric, after 3 weeks salinity treatment at assigned ontogenetic stage (short-term experiment).

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